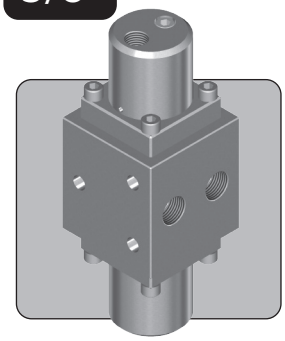
- S-6B37-G -

DIRECTIONAL CONTROL VALVE

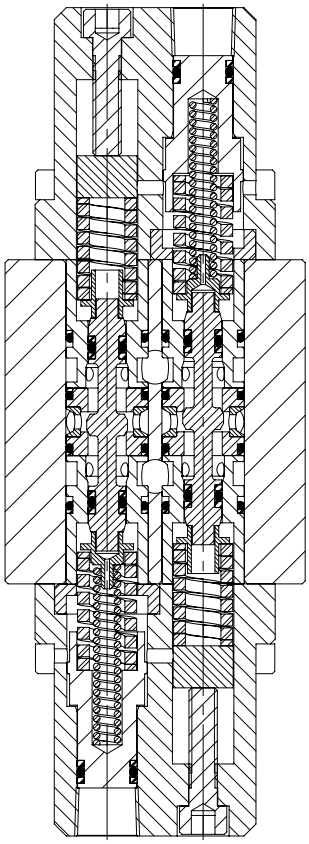
3/8″ UP TO 414 BAR 6,000 PSI



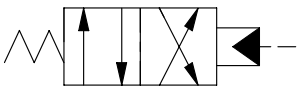
The S-6B37-G is a 3/8″ nominal bore 2 position, 4 way directional control valve which is fitted with 2 hydraulic actuators that are designed to be pressurised simultaneously.

The valve uses a balanced spool and seat design, such that pressure in the main lines does not affect the actuator force required to operate the valve. The valve design incorporates 2 of these balanced spool assemblies. The valve offers very low leakage from its soft seated design.

- 2 position 4 way directional control valve
- Balanced spool design, with spring return
- Soft seat design, offering low leakage for liquid & gas duty
- Suitable for use with mineral oils and water glycols, seawater, air, nitrogen and sweet natural gas
- Suitable for many other media's, contact us for advice
- Vent chambers prevents mixing of main line and pilot fluids
- Versions available for subsea use (contact us for advice)
- Wide pilot pressure range
- Various seal and porting options available



Specifications

BASIC MODEL NUMBER	S-6B37-G
SYMBOL	
MAX WORKING PRESSURE (LIQUID)	414 bar (6,000 psi)
MAX WORKING PRESSURE (GAS)	241 bar (3,500 psi)
CV (FLOW CAPACITY)	0.44
FLUID	Liquids & Gases See materials section
TEMPERATURE RANGE	See Product Selector opposite and Technical Data section
PORT SIZE	3/8″
WEIGHT	4 kg (8.8 lb)

Specifications may change without notice

Materials

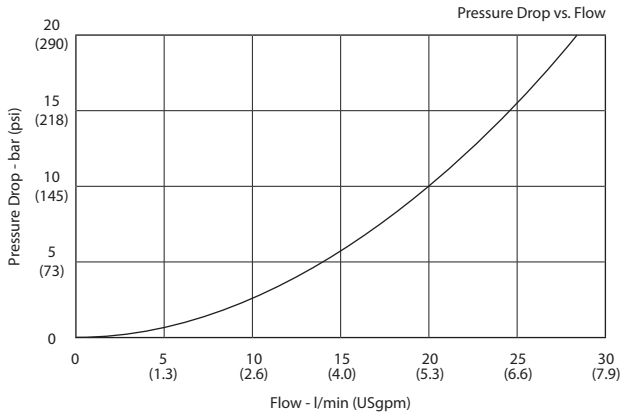
Externally Exposed Parts: 316 Stainless Steel with an electroless nickel plated spring.

Internally Wetted Parts: 316 Stainless Steel, Torlon, PTFE and Acetal

The standard valve is designed for use with air, nitrogen, sweet natural gas, mineral oil, water glycols, plain water and seawater. Contact us for advice when use with other media is required.

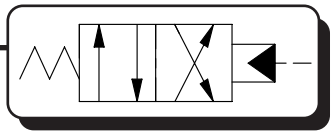
⚠ The standard valve has Viton® seals. Further seal options are available via the Product Selector. Compatibility with the working fluid at the operating temperature must be considered.

Typical Performance



Typical performance based on water

Actuator Options



The S-6B37-G is available with a large number of actuator options. These allow the valve to be configured such that it may be switched manually with a lever, via a cam, or by pilot pressure.

Lever actuators

Two types of lever actuator are available. The L option is a detented lever. Once operated, the lever remains in the selected position. In contrast, the DL option is a deadmans lever. When the lever is released, the spring return will switch the valve and lever back to its default position.

Cam actuators

Two types of cam actuator are available. Both types are isolated from the valve spool by springs, to prevent damage to the spool and seats from excessive mechanical force. It is recommended that some form of adjustment is incorporated into the operating mechanism, to allow for tolerances in the cam actuator.

Pilot actuators

Three types of actuator are available to allow the valve to be switched using pilot pressure. Each actuator type is designed to operate the valve within a particular pilot pressure range. The valve will begin to switch at pressures below the minimum quoted pilot pressure. At the minimum pilot pressure, the valve will be fully switched and will achieve minimum leakage. The valve should not be operated above the maximum pilot pressure as damage to the spool or seats may occur. The table below shows the allowable pilot pressure range for each actuator type.

ACTUATOR TYPE	PILOT PRESSURE RANGE
H	69 to 214 bar (1,000 to 3,100 psi)
A3	4.1 to 10.3 bar (60 to 150 psi)

Note: Interflow exists during switching

Product Selector

S-6B37-G P - H0 - G - NBR - 6K

VALVE TYPE	PORTING OPTIONS	ACTUATOR OPTIONS	VALVE OPTIONS	SEAL OPTIONS	MAX WORKING PRESSURE
S-6B37-G	P 3/8" BSPP female N 3/8" NPT female Further porting options are available on request	H (1,000 to 3,100 psi) A3 (60 to 150 psi)	G option must be selected G Soft seats G option must be selected	Leave blank for Viton® seals -10˚c to +80˚c NBR Nitrile seals -20˚c to +80˚c KLZ Kalrez® seals -20˚c to +80˚c HNBR Hydrogenated Nitrile seals -20˚c to +80˚c Further seal options are available on request	6K 414 bar (6,000 psi) Liquid 241 bar (3,500 psi) Gas

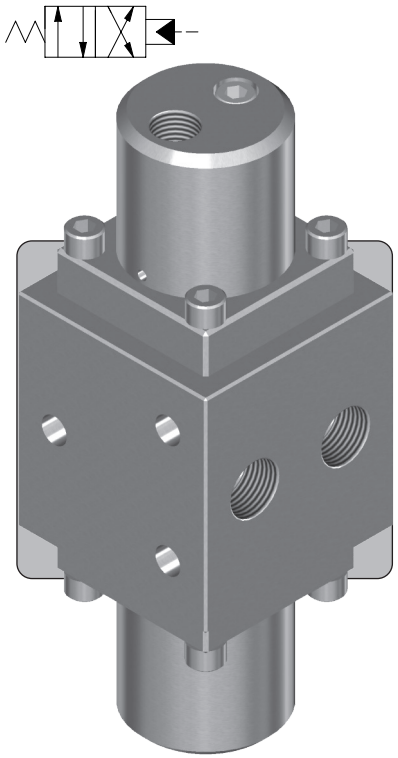
Repair and seal kits are available on request

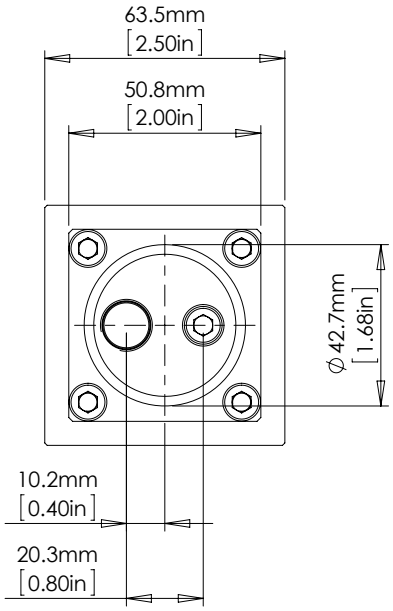
For a seal kit add SK at the end of the model code and for a repair kit add RK

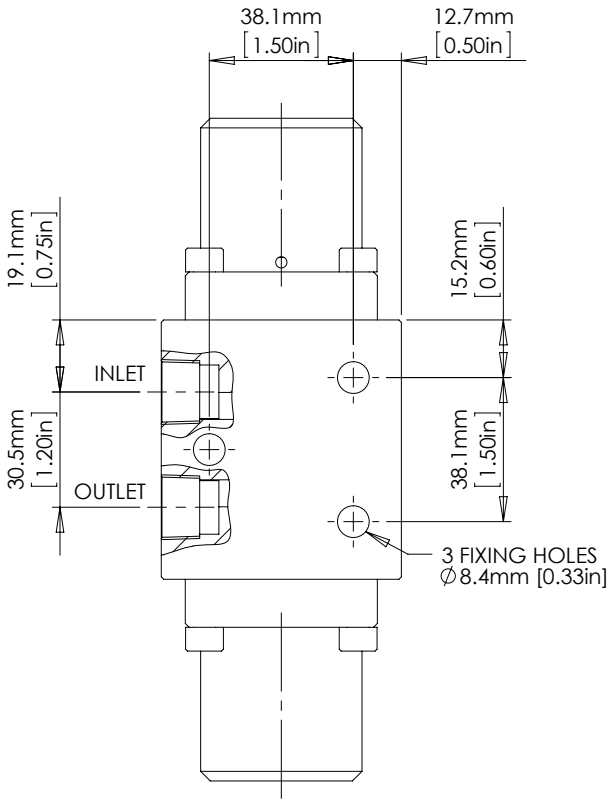
S-6B37-G

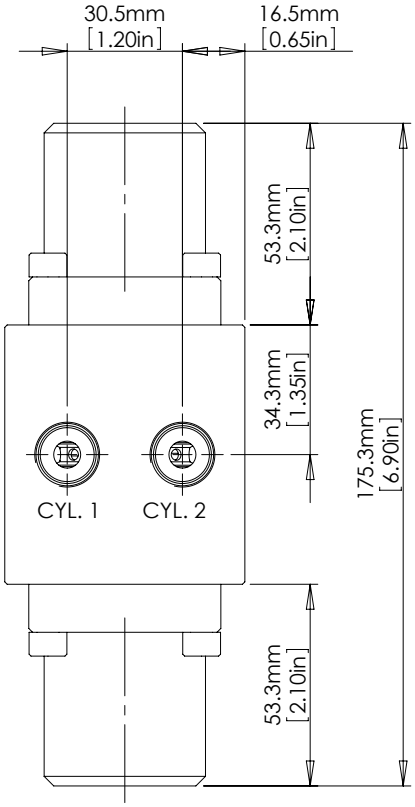
DIRECTIONAL CONTROL VALVE

S-6B37-G-H



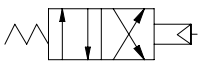


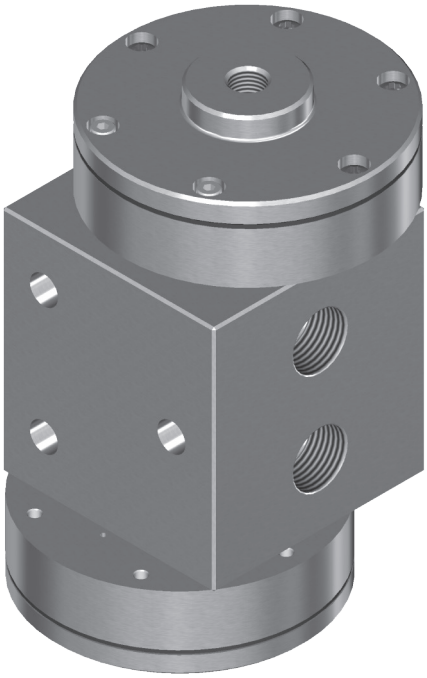


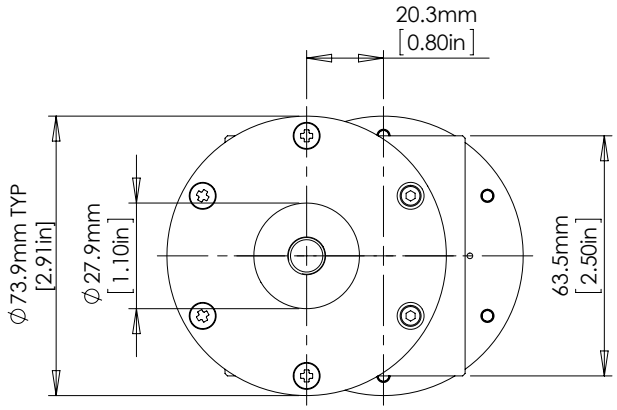


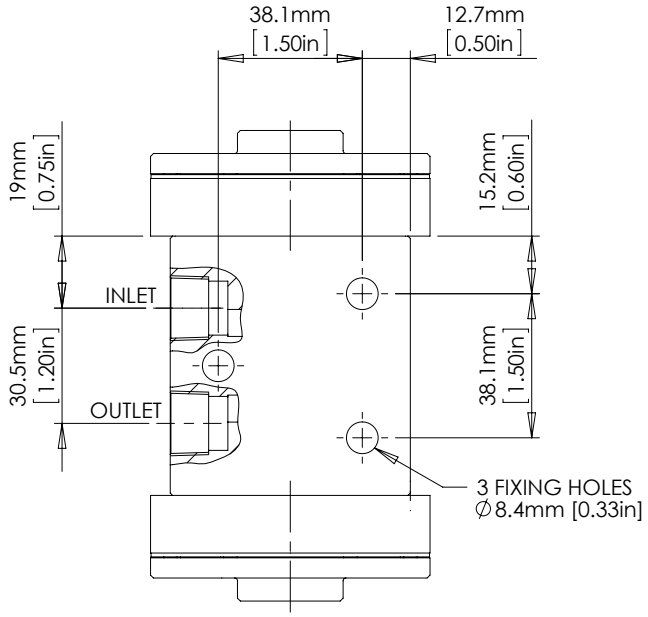
BS VALVES

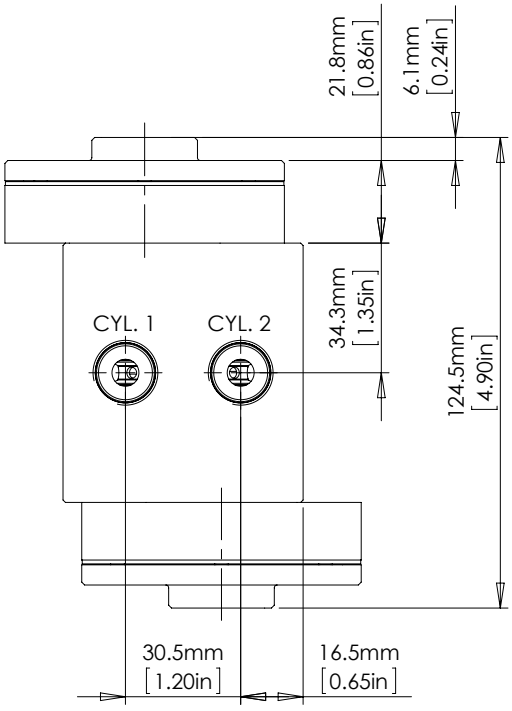
S-6B37-G-A3






20.3mm
[0.80in]
Ø73.9mm TYP
[2.91in]
Ø27.9mm
[1.10in]
63.5mm
[2.50in]


38.1mm
[1.50in]
12.7mm
[0.50in]
19mm
[0.75in]
15.2mm
[0.60in]
INLET
30.5mm
[1.20in]
OUTLET
38.1mm
[1.50in]
3 FIXING HOLES
Ø8.4mm [0.33in]


21.8mm
[0.86in]
6.1mm
[0.24in]
CYL. 1
CYL. 2
34.3mm
[1.35in]
124.5mm
[4.90in]
30.5mm
[1.20in]
16.5mm
[0.65in]